

Tax Disclosures and The Value of Salary Confidentiality^{*}

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Abstract

How much do taxpayers value salary confidentiality? We leverage administrative data to estimate a revealed preference-based sufficient statistic for the willingness to pay for earnings confidentiality. We exploit the fact that nonprofits are required to publicly disclose the names and salaries of certain employees whose taxable earnings exceed \$100,000. We find that taxpayers who bunch below the threshold reduce their pre-tax income by up to \$3,076 to avoid disclosure. Employees whose earnings have recently been disclosed are also more likely to separate from their employers. We argue that this is indicative of an employee demand for privacy.

1 Introduction

Proponents of salary transparency have long argued that greater public access to earnings records can foster accountability and promote public participation in matters of inequality. A recent wave of high-quality studies has demonstrated the potential for salary transparency to reduce pay disparities. However, the desirability of salary transparency mandates depends on the balance of social benefits and private costs imposed on those whose salaries are disclosed. Measures of

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these private costs have proved elusive, as the value of data privacy is idiosyncratic and context-dependent (Acquisti et al., 2016). While some value salary privacy for instrumental purposes, such as the potential effect of disclosure on salary negotiations or social status, others have an intrinsic preference to protect their earnings information from disclosure (Becker, 1980; Lin, 2022). As a result, it is challenging to estimate the effect of tax disclosures on the welfare of workers because of the difficulty of comprehensively measuring the various real effects and underlying preferences that shape the demand for data privacy.

This study makes progress on this challenge by estimating a revealed-preference-based sufficient statistic for the value of salary confidentiality. We do this in the context of the only federal income transparency policy embedded in the U.S. tax code. Our identification strategy exploits the fact that tax-exempt organizations are required to disclose the earnings of the top five highest-paid employees only when their total compensation exceeds \$100,000. This individual salary transparency requirement can be thought of as imposing a notched disclosure-cost function on nonprofit employees.

Due to this transparency policy, a left-censored distribution of nonprofit workers' compensation is publicly visible on nonprofit information returns. We use administrative tax data to construct an uncensored distribution of nonprofit workers' earnings. Using this distribution, we employ a bunching design to estimate these employees' willingness to pay to avoid disclosure. We find that the distribution of pre-tax earnings above the disclosure threshold is distorted up to an earnings level of \$103,076. This translates into a value of salary confidentiality of \$2,215, or approximately 2.2% of pre-tax income.

We match this data to nonprofit employees' confidential tax records, and leverage this matched dataset to investigate how disclosed workers respond to their loss of privacy. We first identify a subset of nonprofits for which the disclosure threshold is difficult to observe, because the fifth-highest-paid non-executive earns well above \$100,000. We verify that there is no evidence that workers manipulate their income to avoid disclosure in this subsample. Thus assured, we examine whether workers are more likely to leave the firm following their first disclosure. We find that disclosure increases employees' separation rate by 2 percentage points in the year following disclosure, and remains elevated for several subsequent years.

Our study helps rationalize the near-ubiquity of restrictive tax privacy policies across jurisdictions. We find that disclosure of information used to calculate individuals' tax liabilities has the

potential to create large distortions to the reporting behavior of taxpayers for reasons unrelated to the tax rate itself. In our context, the design of the disclosure threshold leads to a substantial reduction in reported taxable income when taxpayers can anticipate disclosure, and generates real responses – such as separations – when disclosure is more difficult to anticipate. More generally, insofar as the pressures of disclosure lead taxpayers to alter their reported earnings or labor supply, tax transparency will generally diminish the efficiency of the tax system.

These results are related to the broader literature on salary transparency. Cullen (2024) calculates the implied effect of salary transparency on average wages across three high-quality studies of individual earnings transparency mandates in municipal or university contexts (Baker et al., 2023; Mas, 2017; Obloj and Zenger, 2022). A simple average of the three studies suggests a treatment effect of -1.8% , which is substantially smaller than our estimated value of confidentiality. This comparison suggests that disclosure carries additional costs, above and beyond their expected impact on earnings.

This study contributes to the growing literature on the economics of earnings transparency. The empirical evidence on the effects of individual disclosure policies is mixed. While studies in the context of tax disclosures have found a small positive effect on reported earnings (Bø et al., 2015; Slemrod et al., 2022), the literature on salary transparency has consistently found that individual disclosure mandates reduce wages by a modest amount (Cullen, 2024). Finally, Neisser and Wehrhöfer (2025) show that politicians’ response to disclosure requirements depends on their political affiliation. The variation in these results underscores the context-specificity of the effects of earnings disclosures.

Finally, our results contribute to the literature on optimal privacy guarantees. Governments often provide detailed public statistics calculated from individual tax records. However, a recent literature on differential privacy has emphasized the disclosure risks associated with the production of these statistics (Dwork (2006)). Abowd and Schmutte (2019) argue that optimal differential privacy guarantees ought to balance the private costs of disclosure against the social benefit of greater statistical accuracy. To our knowledge, this paper is the first to offer a credible estimate of the former. Our estimate of the value of confidentiality can be used by statistical agencies to help calibrate data protections in a way that balances the demand for public statistics with the value of data confidentiality.

The paper proceeds as follows. In Section 2 we provide an overview of the institutional context,

including a detailed description of the salary transparency regulations that govern U.S. nonprofits. In section 3, we provide an overview of our structural framework. Section 4 discusses our empirical methodology, which we implement using data described in Section 5. Our results are presented and discussed in Section 6. We conclude in Section 7.

2 Institutional Context

In the United States, nonprofit organizations are required to file an annual information return in order to maintain their tax-exempt status. This return is called the Form 990.¹ Its contents are required to be made available for public inspection.² The IRS facilitates this public inspection by hosting a Tax Exempt Organization Search function on its website, where a nonprofit’s recent Form 990s can be viewed. These returns are also widely available on charity watchdog sites, such as Charity Navigator, and are typically self-published by nonprofits on their own websites. Any element included on a Form 990 is therefore both publicly disclosed and widely disseminated.

Figure 1: Form 990 Part VII

(A) Name and title	(B) Average hours per week (list any hours for related organizations below dotted line)	(C) Position (do not check more than one box, unless person is both an officer and a director/trustee)						(D) Reportable compensation from the organization (W-2/ 1099-MISC/ 1099-NEC)	(E) Reportable compensation from related organizations (W-2/ 1099-MISC/ 1099-NEC)	(F) Estimated amount of other compensation from the organization and related organizations
		Individual trustee or director	Institutional trustee	Officer	Key employee	Highest compensated employee	Former			
(1)										
(2)										
(3)										
(4)										

Crucially, Part VII of the Form 990 requires the nonprofit to list the income of each executive³ and up to five other employees. These salary disclosure requirements are unique, in the sense

¹Only sufficiently large organizations are required to file the Form 990. Since 2008, this form is required for all organizations which earn gross revenues of over \$50,000 or hold assets of over \$500,000. Smaller organizations file simpler versions of the Form 990: either the abbreviated Form 990-EZ, or the Form 990-N postcard return.

²See 26 U.S.C. §6104(b).

³Here, nonprofit executives are defined to include officers, directors, trustees, and key employees.

that they are embedded in the U.S. tax code.⁴ As shown in Figure 1, the names of each of these individuals appear alongside their total reportable compensation (TRC). Further details on this definition of compensation are presented in Section 5.

Nonprofit organizations therefore impose disclosure costs on some of their employees, as a consequence of the organization’s tax status. Officers, directors, trustees, and key employees must pay these disclosure costs regardless of their income. However, the five highest-paid employees outside those categories face a disclosure threshold, located at \$100,000. This threshold creates a notch in the disclosure costs paid by non-executive nonprofit employees. The private cost of salary disclosure – or, equivalently, the value of salary confidentiality – can be estimated by examining patterns in the distribution of nonprofit workers’ earnings around this disclosure threshold, and contextualizing these patterns in a model of the worker’s behavior.

3 Conceptual Framework

In this section, we provide an overview of our model and empirical methodology. Our model taxpayer supplies labor to a tax exempt organization whose disclosure regime is captured by the indicator D . Specifically, $D = 1$ if the worker’s employer is subject to an earnings disclosure rule, and $D = 0$ when they are not. The value of a type $n \sim F(n)$ taxpayer’s choice of salary, conditional on the disclosure regime, reflects the following maximization problem:

$$\begin{aligned} \max_z u(z; n) &= c - \Theta(z) \cdot D - \psi(z; n) \\ \text{subject to } c &\leq z - T(z) \end{aligned} \tag{1}$$

where c is consumption, z is pre-tax earnings, $T(z)$ is the tax function, $\Theta(z)$ is the disclosure costs incurred by taxpayers when compensation is disclosed to the public, and $\psi(z; n)$ represents the utility cost of earning income.

Absent earnings disclosure ($D = 0$), labor supply is chosen to satisfy the following first order condition:

⁴In general, there is no federal legal standard that guarantees salary confidentiality for public or private employers. The salaries of government employees are often public, and data vendors such as Experian collect and sell detailed and identified payroll records, through products like The Work Number.

$$1 - T'(z) - \psi'(z; n) = 0 \quad (2)$$

This equation is an implicit function of n and z that maps the smooth ability distribution $F(n)$ into a smooth distribution of earnings $H_0(z)$. This is the counterfactual earnings distribution that would prevail in a world with no disclosure risk.

The disclosure policy that we consider in our empirical exercise requires the taxpayer's employer to disclose taxable compensation if it exceeds a reporting threshold z^* . Therefore, taxpayers face a fixed private cost associated with such disclosures akin to a notch. As we will see later, this is consistent with the observed pattern of earnings around the disclosure threshold, with sharp bunching exactly below the disclosure threshold and a missing mass to the right. Therefore, our model taxpayer's transparency cost function takes the following form:

$$\Theta(z) = \Theta \cdot \mathbf{I}\{z > z^*\} \quad (3)$$

The value of the parameter Θ captures the direct welfare cost of salary transparency and the value of salary confidentiality. Thus, it is the central object of interest in our analysis. The existence of this notched disclosure cost implies that taxpayers just above the notch will face an incentive to reduce their taxable income by an amount Δz^* in order maintain the salary confidentiality.

The pattern of bunching, which is indicative of the behavioral response to the disclosure notch, is informative about the magnitude of Θ because taxpayers near the kink reduce their earnings to avoid disclosure if the following inequality holds, with equality for the marginal buncher:

$$\Theta \geq \Delta z^* - \Delta T + \Delta \psi \quad (4)$$

where Δz^* is the pre-tax earnings change needed to avoid disclosure, $\Delta T = T(z) - T(z^*)$ is the adjustment for tax liabilities in the bunching region, and $\Delta \psi = \psi(z; n) - \psi(z^*; n)$ captures the nonpecuniary benefits of a reduction in earnings, such as less effort or increased nontaxable and non-reportable perquisites.

Because the last term of the right hand side is not observed directly, we can only estimate the first two terms. However, since its value is strictly positive, the first two terms provide an upper

bound for the value of salary confidentiality.

$$\Theta \leq \Delta z^* - \Delta T \approx \Delta z^* \cdot (1 - t) \quad (5)$$

We argue that equation (5) is also a reasonable approximation of the welfare cost of disclosure. At the extreme, assuming $\Delta\psi = 0$ represents the case where the taxpayer simply takes a pay cut to avoid salary disclosure, without any corresponding changes in work effort or perquisites. To see this, first note that the pay-cut scenario is the most likely to yield a Pareto improvement relative to the earnings level that would have prevailed absent disclosure. This is because the worker gains privacy while the employer saves on labor costs. Second, any perquisites that an employer may provide are likely have only a second order effect on the worker’s welfare. This is because perquisites are already an important part of employee compensation in the nonprofit context, so that the employer has already exhausted many opportunities to substitute earnings for nontaxable fringe benefits. This logic also implies that the provision of perquisites necessary to overcome the welfare effect of lost earnings may be costly to the employer, further reducing the likelihood that perquisites can be used to effectively offset the cost of reduced labor earnings.

4 Empirical Methodology

Section 3 illustrates that the value of salary confidentiality is closely related to the behavioral response to the transparency policy. To estimate this behavioral response, we adapt the estimation algorithm outlined in Kleven and Waseem (2013). This allows us to discover the value nonprofit employees place on avoiding disclosure. This strategy is most useful for estimating this value in nonprofit workplaces where fewer than 5 highly-compensated employees are disclosed, as the \$100,000 disclosure threshold is then observable with certainty. At the remaining nonprofits, employees cannot perfectly observe the disclosure threshold, and so may have difficulty bunching to avoid disclosure.

Although these employees cannot easily avoid disclosure, they can still respond to disclosure after the fact. This provides an opportunity to study the effects of disclosure. We aim to characterize responses to disclosure among employees who cannot avoid it. To do so, we show that in nonprofit workplaces where exactly 5 highly-compensated employees are disclosed, the initial disclosure event occurs as-if randomly. We exploit this discovery to identify the effect of initial

disclosure on newly-disclosed employees' separation rates.

4.1 Identifying the Value of Disclosure Avoidance

As discussed in Section 3, nonprofits must disclose total reportable compensation for the five highest-paid non-executive employees making over \$100,000. This rule implies that the disclosure risk faced by nonprofit employees in the vicinity of the \$100,000 threshold depends crucially on their employer. When a nonprofit has fewer than 5 non-executive employees making over \$100,000, its remaining workers face disclosure at the \$100,000 threshold with certainty. On the other hand, if their employer already discloses the compensation of 5 non-executive employees on Form 990, then the disclosure threshold faced by the sixth-highest-paid of these employees is strictly greater than \$100,000. We therefore expect that employees of nonprofits in the former group – the “high-disclosure-risk” group – should bunch at the \$100,000 threshold, to a greater extent than employees of nonprofits in the latter group – the “low-disclosure-risk” group.

We split the sample according to the disclosure risk group of the nonprofit employer, and proceed to implement the Kleven and Waseem (2013) bunching algorithm separately for each group. This is a method of finding the “bunching window” by requiring that the excess mass below the bunching threshold – in this case, \$100,000 – equal a “missing mass” above the bunching threshold. The lower bound of the bunching window is determined by inspection. The upper bound of the bunching window is the value of total reportable compensation which equates the excess mass with the missing mass.

To determine the excess mass, we must estimate the counterfactual distribution of total reportable compensation in the absence of the notched disclosure function. We estimate this counterfactual using the following specification:

$$\frac{c_j}{N} = \sum_{i=0}^p \beta_i (z_j)^i + \sum_{i=z_L}^{z_U} \gamma_i \mathbf{1}(z_j = i) + \sum_{r \in R} \rho_r \mathbf{1}\left(\frac{z_j}{r} \in \mathbb{N}\right) + \nu_j \quad (6)$$

Here, c_j denotes counts of earnings records in each \$1 earnings bin j , N denotes the overall number of earnings records in the relevant disclosure risk group, p denotes the degree of the polynomial fit, and z_j denotes the earnings bin. We include a set of round-number indicators for multiples of $R = \{1K, 5K, 10K, 25K\}$. Finally, ν_j is an OLS error term. The counterfactual distribution can be recovered by retrieving the predicted values of (6), under the counterfactual

assumption that the parameters $\gamma_i = 0$.

We estimate Equation (6) iteratively, increasing z_U at each step, until the excess mass is equal to the bunching mass. The bunching mass is defined as the difference between the actual distribution and the counterfactual distribution, integrated from the lower bound of the bunching window, z_L , to the disclosure threshold, \$100,000. The excess mass is defined as the difference between the counterfactual distribution and the actual distribution, integrated from the disclosure threshold to the upper bound of the bunching window. Like Kleven and Waseem (2013), we choose a lower bound by visual inspection, setting it at \$99,600.

The behavioral response for the marginal buncher can therefore be estimated as $\Delta z^* = z_U - z^*$. This estimate can be plugged into Equation (5) to produce an estimate of the value of salary confidentiality, given a marginal tax rate.

4.2 Identifying the Response to Disclosure

The strategy laid out in the previous section estimates the value that nonprofit employees place on avoiding disclosure. However, only employees of high-disclosure-risk nonprofits, with less than five disclosed employees, have the ability to observe the bunching threshold with certainty. At other nonprofits, employees will be disclosed only when they become the fifth-highest-paid non-executive. These employees have limited ability to anticipate when, or whether, this will occur. If they value salary confidentiality, they are more likely to respond to disclosure, since their ability to prevent disclosure is limited.

In this paper, we study separations from the firm as a potential response to disclosure. This response can be detected in administrative tax data in a straightforward way. If an employee is observed receiving compensation from a given employer in one year, but not in the subsequent year, we may infer that they have separated from the organization at some point of the first year.

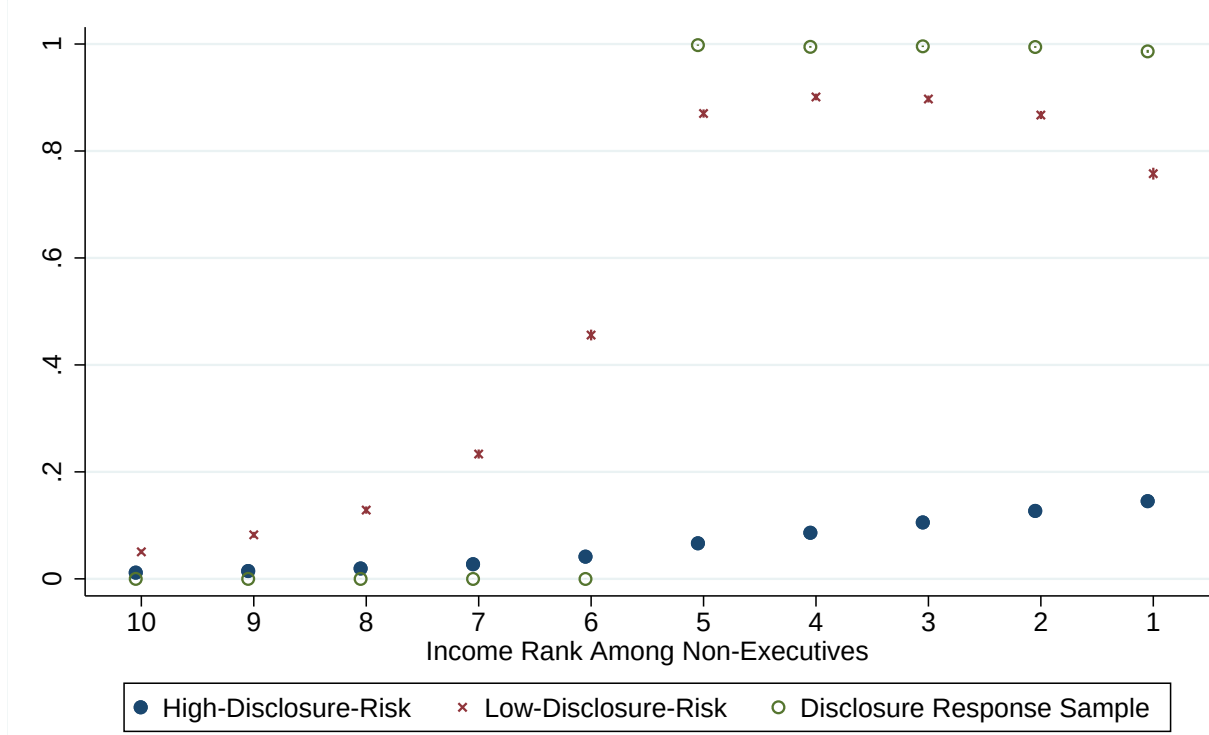
To investigate this hypothesis, we estimate specifications of the following form:

$$Separate_{ifk} = \delta FirstDisclosure_{ift} + \varepsilon_{ift} \quad (7)$$

where $FirstDisclosure_{ift}$ is an indicator, equal to 1 if employee i of focal nonprofit f was first listed on Form 990 Part VII during year t . The outcome variable, $Separate_{ifk}$, is an indicator which takes the value 1 if the employee i does not receive any reportable compensation from focal

nonprofit f in year $k \in \{t+1, t+2, t+3\}$. Finally, X_{ift} represents a vector of controls, which includes employer and year fixed effects.

Figure 2: Probability of a Match between Tax Data and Form 990 Disclosures, by Income Rank and Sample



Notes: This figure plots point estimates and 95% confidence intervals from regression output. The outcome variable is an indicator equal to 1 when an observation from tax data has been successfully matched to a non-executive disclosed on the nonprofit employer’s Form 990 Part VII for the corresponding tax year, indicating disclosure. The left-hand side of the regression includes a series of ten binary variables, which indicate the employee’s rank in the within-firm income distribution for that year. The incomes of nonprofit executives are excluded from this ranking. Nonprofit executives are defined as officers, directors, trustees, and key employees; for further detail on these definitions, see the instructions to Form 990. Matches to the “high-disclosure-risk” sample are marked in blue; these workplaces disclose strictly less than 5 non-executives. Matches to the “low-disclosure-risk” sample are marked in red; these workplaces disclose exactly 5 non-executives. Matches to the disclosure response sample are marked in green; this is the subset of the “low-disclosure-risk” sample which purges measurement error to the greatest extent possible. Match procedures are described in Appendix A.

OLS estimates of (7) could suffer from two forms of bias. The first may arise from measurement error in $FirstDisclosure_{ift}$, which would attenuate the estimate of δ towards zero. This measurement error could derive either from an error in the matching process, which is detailed in Appendix

A, or from a data entry error on the part of the nonprofit. To minimize measurement error among our estimation sample, we restrict our sample to include only low-disclosure-risk nonprofits where the fifth-highest-paid non-executive observed in the administrative data matches to the lowest-paid non-executive we observe on the Form 990 Part VII. Match rates among this subsample are presented in Figure 2, along with the corresponding match rates for all “low-disclosure-risk” and “high-disclosure-risk” nonprofit-years in the dataset. By restricting our attention to this estimation subsample, we effectively eliminate any attenuation bias in (7) attributable to measurement error.

The second form of bias occurs because both disclosure, and the outcome variable, depend on income. If an individual’s income affects the likelihood that they leave their employer, independently of the effect of income on disclosure, then this relationship will generate omitted variable bias in (7) unless a control for income is included. We would therefore like to include a control for income – as measured by the natural log of total reportable compensation – in X_{ift} , provided that this variable proves exogenous in this subsample.

In order to assign a causal interpretation to δ in Specification (7), we must therefore impose an assumption that employees of “low-disclosure-risk” nonprofits cannot manipulate their incomes to avoid disclosure. This assumption plausibly holds in the “low-disclosure-risk” sample, where exactly five non-executives are disclosed on 990 Part VII, because the de-facto disclosure threshold for the “low-disclosure-risk” group varies over time. In order to successfully manipulate their income to avoid disclosure, the employee would first have to correctly predict the location of the disclosure threshold in their workplace in the current year. Under this identifying assumption, for those who have not previously been disclosed, disclosure occurs as-if randomly.

In Section 6.2, we validate this assumption in two ways. First, we look for evidence that employees bunch just below the disclosure threshold. In this sample, this threshold varies at the nonprofit-year level. We therefore normalize total reportable compensation (TRC) within the nonprofit-year by the total reportable compensation of the fifth-highest-paid non-executive, and look for bunching just below that threshold. Second, we consider the possibility that employees ranked lower than fifth in the firm’s income distribution manage to avoid disclosure by receiving smaller raises as they near the disclosure threshold. If there are no large, discontinuous changes in TRC – either in level terms within a period, or in percentage changes relative to the prior period – as normalized TRC approaches 0, then this provides strong evidence that our identifying

assumption holds, so that income is plausibly exogenous at first disclosure.⁵

5 Data

The information disclosed on Form 990 Part VII is both publicly available and widely distributed. However, when examined on its own, this table cannot shed much light on the value nonprofit employees place on the privacy of their tax information. Bunching designs, such as the one described in the previous section, require the researcher to observe the distribution of total reportable compensation below the disclosure threshold, as well as above it. Fortunately, total reportable compensation is well-defined by the IRS, and each of its component parts are observable in tax data. This section describes how we construct the full distribution of this variable using administrative tax data.

We begin with the universe of organizations filing Form 990 electronically between tax years 2011 and 2018, inclusive.⁶ Any individual who receives a Form W-2 or 1099-MISC from one of these employers is defined as a nonprofit employee.⁷ Our goal is to retrieve data on any income these individuals receive from their employers which would be included in the definition of total reportable compensation.

The IRS defines total reportable compensation as the sum of non-employee compensation,⁸ medical and healthcare payments,⁹ and the greater of wages, tips, and other compensation or Medicare wages and tips.¹⁰ This income may be paid either by the nonprofit itself – the “focal organization” – or it may be paid by organizations related to that nonprofit. Related organizations are identified using the focal organization’s Form 990 Schedule R.

Following the formula laid out in the Form 990 instructions, total reportable compensation is

⁵If a worker’s income is plausibly exogenous in absolute terms, it follows that it will be plausibly exogenous in relative terms too. The variable $FirstDisclosure_{ift}$ depends both on a worker’s own income and on the full income distribution within the firm. We assume that if a nonprofit non-executive cannot manipulate their own income, they certainly cannot manipulate the incomes of their coworkers.

⁶The focus on electronic filings is motivated by the availability of digitized information for Form 990 Part VII and Form 990 Schedule R: at time of writing, this information has been digitalized for electronic filers only.

⁷If a nonprofit lists a different EIN as the payer on its employees’ Forms W-2 or 1099, the employees of that nonprofit may fail to enter our dataset. However, this measurement error would only affect the internal validity of these estimates if nonprofits were systematically choosing to pay employees to one side of the disclosure threshold under a different EIN than those on the opposite side of the threshold. We consider this unlikely.

⁸Prior to 2019, this information was found in Box 7 of Form 1099-MISC. From 2019 onwards, this information can be found in Box 1 of Form 1099-NEC.

⁹Form 1099-MISC, Box 6.

¹⁰Boxes 1 and 5 of Form W-2, respectively.

calculated as reportable compensation from the focal organization, plus reportable compensation from related organizations, provided that the latter exceeds \$10,000. This formula is represented in Equation (8), where i indexes the employee and t represents the tax year. Focal nonprofits are denoted f ; each focal nonprofit has a possibly empty set of related organizations, $R(f)$.

$$TRC_{ift} = RC_{ift} + \sum_{r \in R(f)} \max\{RC_{irt} - 10,000, 0\} \quad (8)$$

We are interested in characterizing the size of the bunch in TRC at the \$100,000 disclosure threshold, and finding the upper bound of the associated bunching window. This requires estimation of the counterfactual distribution of TRC around \$100,000, if this value did not correspond to a notch in disclosure costs. Due to the local nature of the bunching strategy, estimation of this counterfactual relies on observations relatively close to this threshold. Specifically, we use the portion of the distribution of TRC which ranges from \$75,000 to \$125,000, inclusive.

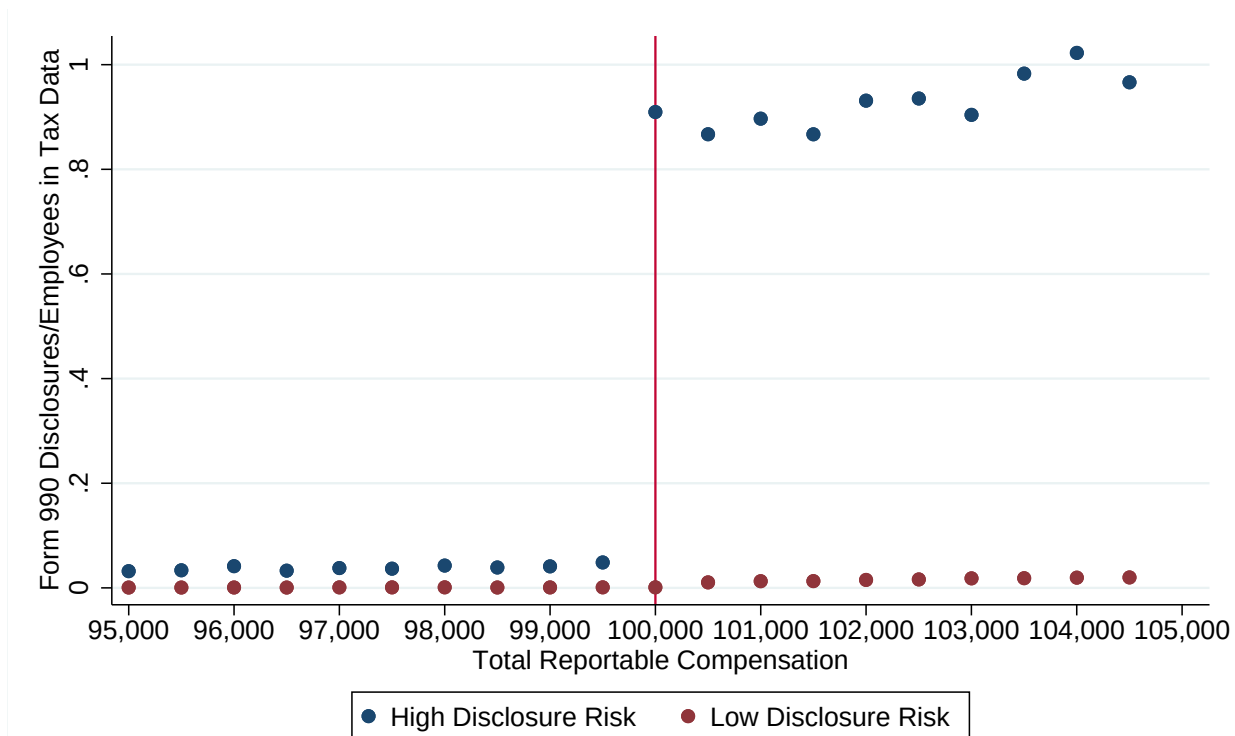
One further restriction on the data arises from the fact that a single individual may be paid by multiple organizations in a given tax year. To see how, consider a case where two nonprofits pay one individual. Suppose organization A pays this person \$100,000, while organization B pays that person \$5,000. If organizations A and B are related to one another, then this individual would receive TRC of \$100,000 from organization A, but TRC of \$95,000 from organization B. That is, this individual would be double-counted. This double-counting is likely to inflate the size of the bunch, especially if individuals paid by multiple related organizations are more aware of disclosure rules than other nonprofit non-executives. We therefore drop all individuals who appear more than once in the relevant domain of the distribution of TRC.

As discussed in Section 3, not all nonprofit employees face a disclosure threshold at \$100,000. This threshold is faced only by non-executives employed at high-disclosure-risk organizations, where fewer than 5 non-executives receive TRC above \$100,000. We therefore proceed to identify organizations by the degree of disclosure risk their employees face at this threshold, and to identify individuals as executives or non-executives within their organizations.

To associate individuals with their roles within the organization, we fuzzy-match their names and total reportable compensation as observed in the administrative tax data with their names and total reportable compensation as observed on Form 990 Part VII. Further details of this matching procedure are described in Appendix A. We remove individuals from the main estimation sample

when they are confirmed to be executives within their organization,¹¹ as these employees are always disclosed, regardless of how much reportable compensation they receive.

Figure 3: Share of Disclosed Individuals, by Income and Employer Type



Notes: This figure plots the number of non-executives listed on a nonprofit’s Form 990 Part VII in a given year, as a share of the number of nonprofit employees found in administrative tax data with calculated total reportable compensation in the corresponding \$500 bin. These shares are presented separately by type of employer. Employers are considered “high-disclosure-risk” if they disclose strictly fewer than 5 employees on Form 990 Part VII. This implies that employees face a high risk of disclosure at the \$100,000 threshold. Employers are considered “low-disclosure-risk” if they disclose exactly 5 employees on Form 990 Part VII. This raises the de facto disclosure threshold for employees at these organizations above \$100,000, such that they face a low risk of disclosure at that \$100,000 threshold.

To determine the degree of an organization’s disclosure risk, we match each focal organization-tax year tuple (f, t) to its Form 990 Part VII filing. We use Column C of that table to count the number of “highest compensated employees” listed by that nonprofit in that year. If the number is less than 5, we consider the nonprofit a high-disclosure-risk employer in that tax year: crossing the \$100,000 threshold would trigger disclosure. If that number is equal to 5, the nonprofit is

¹¹This occurs if the individual is matched to the Part VII table, but they are not listed in that table as a “highest compensated employee”, per Column C.

classed as a low-disclosure-risk employer for that tax year.¹²

Do workers at the charities assigned to the “high disclosure risk” category actually face a higher risk of disclosure at the \$100,000 threshold than their counterparts assigned to the “low disclosure risk” category? To answer this question, we count the number of disclosed non-executives listed on Form 990 Part VII within a given \$500 bin of TRC. We then compare it to the number of non-executive employees within our estimation sample, whose calculated TRC, based on tax data, is located in the comparable \$500 bin. Figure 3 plots the ratio of the number of non-executives listed on Form 990 Part VII to the count of non-executives discoverable in administrative tax data, separately by disclosure risk category. As TRC rises past the \$100,000 threshold, the share of disclosed individuals increases by over 80 percentage points across the threshold. The share of disclosed non-executives hovers near zero for TRC bins below \$100,000, and ranges between 85% to just over 100% for TRC bins above \$100,000.¹³

Note that Figure 3 represents the probability of disclosure in each bin, without imposing the condition that each observation in the Form 990 data can be matched to the corresponding observation in tax data. Appendix A presents a variant of Figure 3, which plots the share of observations within each bin of TRC, calculated from tax data, which are successfully matched to an entry on the Form 990 Part VII.

Table 1 shows the mean value of nonprofit-level covariates, first for high-disclosure-risk nonprofits, and then for the full sample. High-disclosure-risk nonprofits are smaller than average, both in terms of finances and number of personnel. This difference is to be expected, since disclosure risk falls mechanically as nonprofits hire more non-executive employees at six-figure salaries. It follows that low-disclosure-risk nonprofits out-earn and out-spend high-disclosure-risk nonprofits. The smaller nonprofits are somewhat more reliant on revenue from contributions, and are less likely to support themselves with revenue from program services, compared to the average nonprofit. High-disclosure-risk nonprofits appear to spend their money in similar ways to low-disclosure-risk nonprofits: the program service expense ratio, administrative expense ratio, and the share of expenses devoted to payroll all look broadly similar across groups.

¹²In some cases, nonprofits listed more than 5 “highest compensated employees;” these organizations were discarded from the data for non-adherence to the Form 990 instructions.

¹³Values in excess of 100% may occur because the denominator omits double-counted individuals, whereas the numerator does not. Another source of measurement error in this figure may arise if the nonprofit misreports TRC for some employee; this could result in an employee falling into a different TRC bin in tax data compared to Form 990 data, or in an extreme case, not appearing on Form 990 at all.

Table 1: Mean Characteristics of Nonprofit-Years by Employees' Disclosure Risk

	High Disclosure Risk	Low Disclosure Risk
Total Revenue	20,168.10	198,472.21
% From Contributions	28.08	17.29
% From Program Services	62.98	75.65
Total Expenses	18,996.23	188,261.40
Program Services Expense Ratio	65.57	71.28
Administrative Expense Ratio	12.20	12.31
% Compensation	47.68	47.61
% to Executives	14.84	9.17
% to Non-Executives	79.04	85.13
Employment (Form 990)	246.10	1,303.32
Total Assets	44,063.50	374,735.58
Total Liabilities	21,604.66	184,879.70
# Employee-Years	517,767	7,099,541
# Nonprofit-Years	45,050	48,568
# Nonprofits	13,104	9,764

Notes. This table presents means of nonprofit-year level covariates. Nonprofit-years are classified as high-disclosure-risk if the nonprofit listed less than 5 non-executives on Form 990 Part VII in that year. The full sample includes both high-disclosure-risk and low-disclosure-risk nonprofits. Nonprofit-years belong to the latter category if the nonprofit listed exactly 5 non-executives on Form 990 Part VII in that year. Statistics for these groups are presented in Columns 1 and 2, respectively. All financial variables are presented in thousands of current USD. Share and percentage variables are scaled between 0 and 100. This table includes statistics for nonprofits where at least one employee receives total reportable compensation of between \$75,000 and \$125,000. The count of employee-years reflects only the observations used in the bunching estimation: it excludes officers and any individual who receives total reportable compensation of between \$75,000 and \$125,000 from two distinct organizations in the same year.

6 Results

In Section 6.1, we present evidence that when nonprofit employees can easily observe the disclosure threshold, they bunch to avoid disclosure. In Section 6.2, we show that when the exact location of the disclosure threshold cannot be observed with certainty, an initial disclosure increases the probability that a nonprofit employee separates from the firm in subsequent years. Both sets of results constitute clear evidence that nonprofit employees value the confidentiality of their earnings records. Section 6.3 offers discussion of these results in the context of the broader literature.

6.1 Nonprofit Employees Bunch to Avoid Disclosure

In this section, we present evidence of privacy-motivated bunching at the \$100,000 threshold. We demonstrate that there exists a “missing mass” in the distribution of reportable compensation among employees of high-disclosure-risk nonprofits, but not among employees of low-disclosure-risk nonprofits. We estimate the upper bound of this missing mass, and use it to pin down the implied value of salary privacy to the average nonprofit employee.

Figure 4 illustrates the extent of bunching around the \$100,000 threshold.¹⁴ For employees of high-disclosure-risk nonprofits only, this threshold represents the disclosure notch. Their response is depicted in Figure 4a, whereas bunching by employees of low-disclosure-risk nonprofits is presented in Figure 4b. Employees of high-risk nonprofits exhibit a bunching mass equal to 0.06% of total employment, whereas employees of low-risk nonprofits exhibit a bunching mass of 0.01% of employment. The difference in the bunch is statistically significant.

We find that the upper bound of the bunching window is \$103,076¹⁵ among the high-risk group, implying that the marginal buncher reduces their pre-tax income by up to \$3,076 to avoid disclosure.¹⁶ For the majority of the sample period, an income of approximately \$100,000 would place an unmarried filer into the 28% tax bracket.¹⁷ Plugging this marginal tax rate into Equation (5) yields an implied value of salary confidentiality of \$2,214.72.¹⁸

¹⁴These figures pool across all sample years; the counterfactual density is calculated using a fifth-degree polynomial of total reportable compensation. Appendix B presents alternative specifications, including those with alternative values of p , and which estimate the bunch separately by year. The results are qualitatively similar.

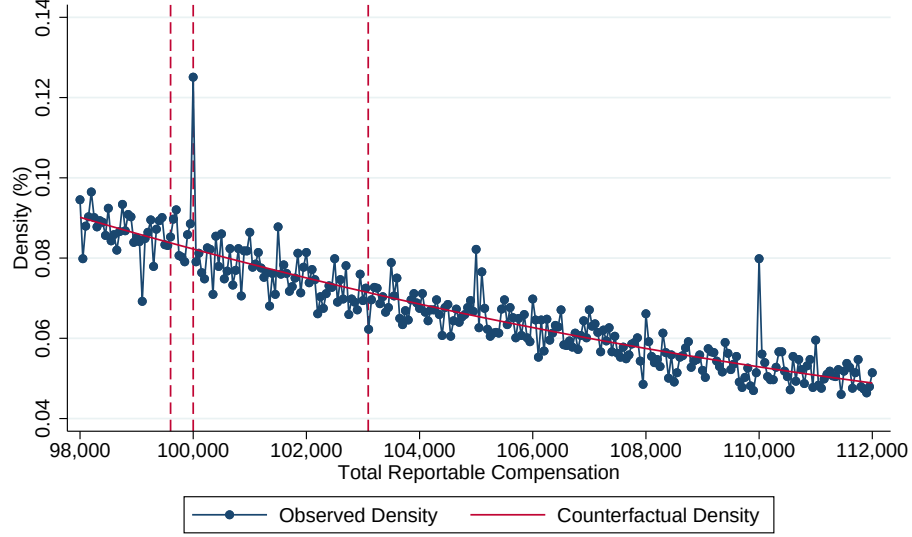
¹⁵Bias-corrected 95% confidence interval: (\$100,731, \$113,787).

¹⁶The upper bound of the bunching window is \$110,533 among the low-risk group, but this high figure is an artifact of the lack of missing mass among individuals at low risk for disclosure. Without a missing mass, the bunching algorithm continues to search for an upper bound over a larger subset of the support of the distribution.

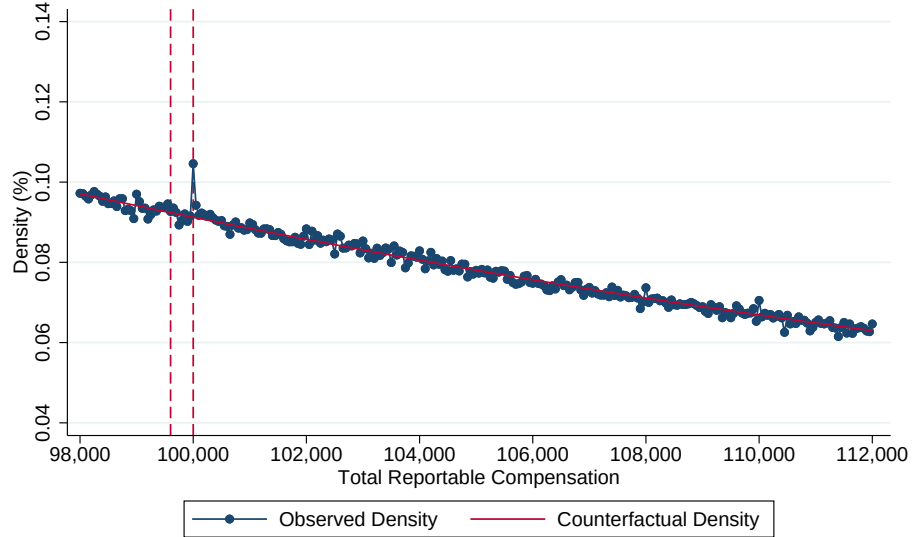
¹⁷In 2018, such a filer would have faced a 24% marginal tax rate.

¹⁸Using the delta method, the bias-corrected 95% confidence interval is given by (\$526.32, \$9,926.64).

Figure 4: Empirical and Counterfactual Densities Around the Disclosure Notch



(a) High Disclosure Risk



(b) Low Disclosure Risk

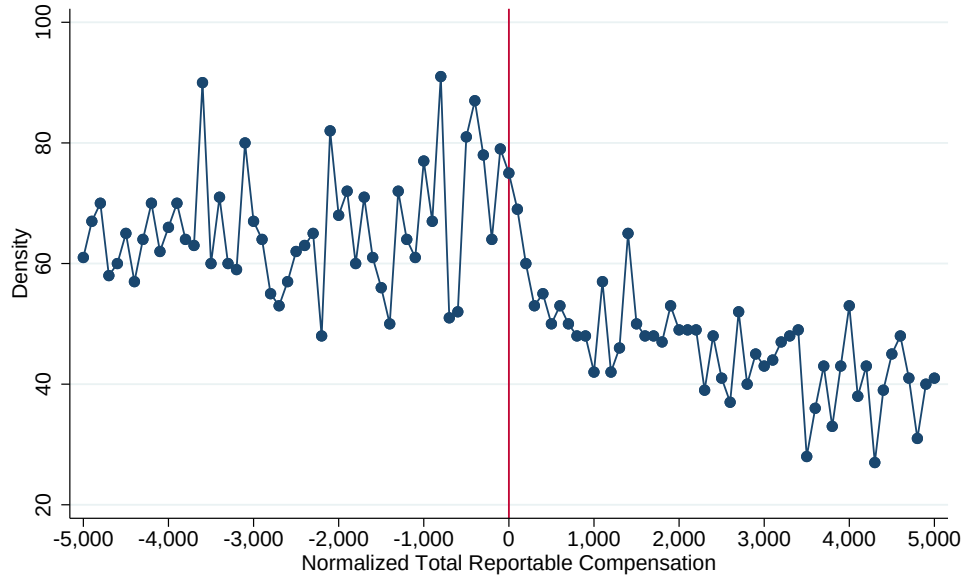
Notes: These figures show the observed and counterfactual densities in the neighborhood of the disclosure threshold, as a function of total reportable compensation in current US dollars. Figure 4a shows the actual and counterfactual densities for employees at high-disclosure-risk nonprofits, which have fewer than 5 non-executives listed on Form 990 Part VII. Figure 4b presents these densities for low-disclosure-risk nonprofits, which have exactly 5 non-executives listed on Form 990 Part VII. In each figure, the three red vertical lines denote the lower bound of the excluded range, the disclosure threshold, and the upper bound of the excluded range, respectively. Each point represents a bin with length \$50. Employees of high-risk nonprofits exhibit a bunching mass equal to 0.06% of total employment (bias-corrected 95% confidence interval: (0.03854%, 0.08664%)). Employees of low-risk nonprofits exhibit a bunching mass of 0.01% of total employment (bias-corrected 95% confidence interval: (0.00455%, 0.01688%)). Confidence intervals are based on a clustered bootstrap resampling procedure with $B = 500$ bootstrap replicates.

6.2 Nonprofit Employees More Likely to Separate after Disclosure

If employees derive disutility from disclosure, how do they respond to disclosure when it occurs? This section asks whether nonprofit employees are more likely to leave their employer after experiencing disclosure on Form 990 Part VII for the first time.

We begin by confirming our identifying assumption that employees of low-disclosure-risk nonprofits do not manipulate their total reportable compensation in order to avoid disclosure. We conjecture that these employees behave differently than those at high-disclosure-risk firms because they have more difficulty observing the disclosure threshold.

Figure 5: Normalized Total Reportable Compensation Smooth Near Zero



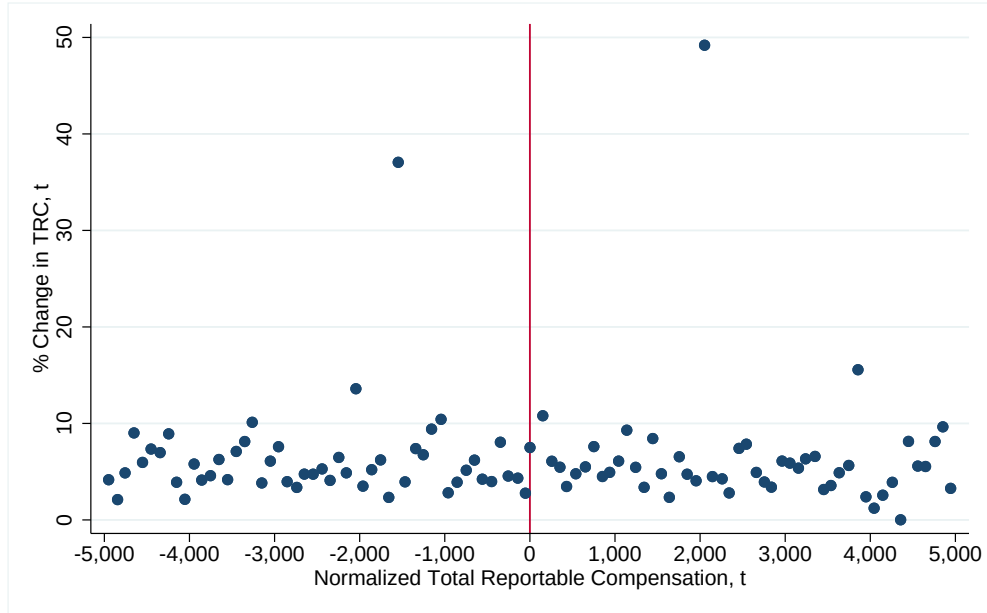
Notes: Each point represents a \$100 bin of normalized total reportable compensation. Normalized total reportable compensation equals \$0 for the fifth-highest-paid non-executive within the organization-year. Observations belonging to the fifth-highest-paid non-executives are omitted from this plot for visual clarity.

The distribution of normalized TRC is presented in Figure 5. The raw data will contain a spike at \$0 by construction;¹⁹ therefore, observations belonging to the fifth-highest-paid non-executives themselves are omitted for visual clarity. This figure shows no evidence of asymmetric bunching in the distribution of TRC around the threshold. While there does appear to be less mass above

¹⁹This occurs because, by definition, every low-disclosure-risk nonprofit has a fifth-highest-paid non-executive. That individual's normalized TRC is equal to zero, again by definition. While each of these organizations will have exactly four employees whose normalized TRC weakly exceeds zero, those employees' salaries are not restricted to be particularly close to zero. Each of these organizations may have any number of employees with normalized TRC strictly less than zero. Again, those employees' salaries are not restricted to lie particularly close to zero.

zero, this is by construction: there must be exactly four employees per nonprofit-year to the right of the threshold, whereas no such quantity restriction applies to the left of the threshold. Figure 6 further illustrates that percentage changes in TRC evolve fairly smoothly as normalized TRC approaches zero. The patterns presented in Figures 5 and 6 therefore support our identifying assumption that employees of “low-disclosure-risk” nonprofits do not manipulate their incomes to avoid disclosure. This permits us to interpret estimates of δ from Specification (7) as the causal impact of an initial disclosure on the probability an employee leaves their employer in subsequent years.

Figure 6: No Systematic Relationship Between Percentage Change in Total Reportable Compensation and Status as Fifth-Ranked Non-Executive



Notes: Each point represents the mean percentage change in total reportable compensation among employees of low-disclosure-risk nonprofits, within a \$100 bin of normalized total reportable compensation. Normalized total reportable compensation equals \$0 for the fifth-highest-paid non-executive within the organization-year.

Table 2 presents the estimates of Specification (7), which characterizes the probability that newly-disclosed employees separate from the nonprofit in each of the three subsequent years. Within each panel, the column number reflects the number of years between the initial disclosure and the timing of the outcome. That is, the first column of Table 2 estimates the probability that disclosure in year t causes the disclosed employee to separate from their job in the subsequent year. The second column models the probability that disclosure in year t causes a separation two

years into the future, and restricts the estimation sample to include only employees who remained at the nonprofit in year $t + 1$. Similarly, the third column models the probability that disclosure in year t leads to a separation in year $t + 3$, conditional on continuing the employment relationship through year $t + 2$.

Table 2: Nonprofit Employees' Responses to Disclosure

	(1)	(2)	(3)
<i>Pr(Separation) in:</i>	$t + 1$	$t + 2$	$t + 3$
First Disclosure, t	0.020*** (0.006)	0.031*** (0.008)	0.056*** (0.011)
$\ln(TRC), t$	-0.048*** (0.001)	-0.044*** (0.002)	-0.052*** (0.004)
Observations	3,243,819	1,571,866	690,622
No. Focal Charities	2,809	2,045	1,344
Remained in $t+1$		X	X
Remained in $t+2$			X
Focal Charity FE	X	X	X
Year FE	X	X	X
Baseline Separations Rate	0.417	0.436	0.385

Notes. This table presents estimates of Specification (7). The outcome variable is an indicator for separation from the nonprofit, measured as of time $t + k$, where t is the year in which each individual first appeared on their organization's Form 990 Part VII filing. All specifications include employer and year fixed effects, along with a control for the natural log of current-period income. Standard errors are clustered at the employer level. Specifications presented in Column 2 are estimated using a subset of observations in Column 1; this sample conditions on an individual remaining at their employer in $t + 1$. Similarly, results presented in Column 3 are estimated conditional on an individual remaining at their employer between period t and $t + 2$. The sample is restricted to the subset of nonprofit-years in which at least one non-executive experiences disclosure on Form 990 Part VII for the first time.

On average, disclosure raises the probability of separation in the following year by 2 percentage points. This difference is both statistically significant and economically important, as it represents

a 4.8% increase in the probability of separation over the mean. The probability of separating from the organization increases in the next two years following the initial disclosure, in both absolute and relative terms.

To understand why the probability of separation increases over time after disclosure, it is helpful to look at the way disclosure evolves over time. Table 3 presents a transition matrix, depicting the rates at which non-executives transition into, and out of, the disclosable set of top-five earners. This transition matrix is calculated conditional on remaining at the current employer. Within that employment relationship, the probability of moving into the “top five” is quite low – only 0.19% – but once there, 83.78% of employees who stay with the nonprofit remain in the disclosable set.

Table 3: Transitions Into and Out Of the Disclosable Set

	Non-Top 5, $t + 1$	Top 5, $t + 1$
Non-Top 5, t	99.81	0.19
Top 5, t	16.22	83.78

Notes. Transition rates are calculated for non-executives who remain with their time- t employer in time $t + 1$. Rates are presented as percentages. The disclosable set includes the top five highest-paid non-executives within a nonprofit-year, where pay is measured by total reportable compensation.

6.3 Discussion

The results of Sections 6.1 and 6.2 are consistent with our hypothesis that nonprofit workers value salary confidentiality. In this section, we discuss the underlying behavioral mechanisms that may manifest in a desire for salary confidentiality. We also examine context-specific factors that might lead to differences in demand for salary confidentiality across sectors.

The academic literature on data privacy suggests that there are at least two reasons why taxpayers might value salary confidentiality. The first is the intrinsic value of privacy, whereby an individual’s wish to remain undisclosed reflects a fundamental preference for discretion or control over their personal information. Another potential source of demand for salary confidentiality is the functional or instrumental value of privacy, whereby taxpayers associate earnings transparency with a perceived negative impact on economic outcomes. This could capture the positive or negative effects of salary transparency (Cullen (2024)) or the impact of disclosing personal information

on price discrimination and other labor market outcomes (Posner (1981)). From the perspective of consumer theory, this distinction can be thought of as differentiating between privacy as a final or intermediate good (Farrell (2012); Becker (1980)). Our estimated willingness to pay for salary transparency reflects a combination of these two motives. It is worth noting, however, that the distinction is not innocuous. As discussed in Lin (2022), the relative prevalence of intrinsic versus instrumental preference for privacy can significantly affect an observer’s perception about whether an undisclosed earner has “something to hide.”

We have shown that nonprofit workers who earn close to \$100,000 value the confidentiality of their taxable earnings as much as \$2,215. This interpretation of the bunching estimate relies on a model in which the worker values the privacy of their own salary, but the employer is indifferent to this concern. How reliable is this assumption of indifference? In theory, nonprofit employers could prefer that their employees stay below the disclosure threshold. After all, if too many employees earn over \$100,000, the nonprofit’s donors may suspect that the organization is inefficient. Increases in labor costs can dampen charitable contributions, due to donors’ well-documented “overhead aversion” (Hung et al. (2023)). Indeed, salient disclosures of nonprofit executive pay have been shown to reduce donations (Balsam and Harris (2014)) and slow further growth of executive compensation (Galle and Walker (2016)).

The results of our second identification strategy can help shed some light on the plausibility of the assumption that our estimated value of confidentiality primarily reflects the worker’s preferences. Suppose by contradiction that only the employers value privacy, and consider a pair of nonprofit coworkers, A and B, who earn similar amounts. As of period t , A is the fifth-highest-paid non-executive in their workplace, and consequently is listed on Form 990 Part VII. As of period t , B is ranked sixth, and has never been disclosed before. B then receives a promotion that causes her to narrowly surpass A in the within-firm income distribution. Unless there is a compelling reason that the employer values employee A’s privacy differently from that of B, there should be no difference in the disclosure costs incurred by the employer because B is disclosed in period $t + 1$ instead of A. In both periods, t and $t + 1$, five non-executives have been disclosed; only the identities of the disclosed employees differ. The aggregate amount of total reportable compensation may increase somewhat due to B’s promotion. However, by revealed preference, the employer would not have promoted B if they did not consider any associated disclosure costs worthwhile. On the other hand, the costs of disclosure are novel for nonprofit employee B, who faces disclosure for the

first time. It follows that if we observe a large reaction to disclosure, this reaction should reflect actions taken by the party which experienced the larger change in disclosure costs. Since we find elevated separation rates following the initial disclosure, it follows that the separation behavior we observe is more likely to reflect the preferences of the employee, rather than the employer.

The external validity of this estimate of the value of privacy may be limited, depending on the relative value that nonprofit and for-profit workers place on the confidentiality of compensation. A long literature posits that nonprofit workers’ motivations differ from those of for-profit workers. For example, nonprofit workers may derive warm-glow utility from “doing good” for a living (Preston (1989), Rose-Ackerman (1996), Glaeser and Shleifer (2001)).²⁰ If nonprofit workers are particularly willing to forgo financial compensation in favor of non-financial compensation, then nonprofit workers may be more willing to bunch below a disclosure threshold compared to workers in the for-profit sector. If this were to hold, then the estimate of Θ produced on a sample of nonprofit employees would overstate the value of privacy among workers in general.

7 Conclusion

This paper investigates the effects of the only salary transparency policy embedded in the U.S. tax code. This policy requires non-profit entities to publicly disclose the compensation of some, but not all, of their employees. Our identification strategy exploits the fact that nonprofits are required to disclose the earnings of the top 5 highest-paid non-executives only when their total compensation exceeds \$100,000. Our analysis relies on the confidential tax records of individual nonprofit employees, matched to public data from their employers’ information returns.

The design of this disclosure threshold implies that there are two types of nonprofits: those for which the \$100,000 disclosure threshold binds, and those for which it does not. The former group discloses fewer than 5 non-executives on its information return, and so when an employee reaches the \$100,000 threshold, they know they will face disclosure. The remaining nonprofits already disclose 5 non-executives, which implies that the disclosure threshold is a moving target, corresponding to the earnings of the 5th-highest-paid individual.

We therefore employ a pair of identification strategies to characterize the value which nonprofit employees place on the privacy of their earnings. When nonprofit employees face the \$100,000 dis-

²⁰This “warm glow” is uncorrelated with pecuniary compensation.

closure with certainty, we find evidence of privacy-induced bunching below that salary transparency threshold. We implement the bunching algorithm of [Kleven and Waseem \(2013\)](#) to estimate that the marginal buncher reduces their pre-tax income by up to \$3,076 to avoid disclosure. Given the corresponding marginal tax rate structure in place during the sample period, this implies that nonprofit employees value their salary confidentiality at \$2,215.

At nonprofits which already disclose the earnings of the 5 highest-paid non-executives, we employ an instrumental-variables strategy to discover how recently-disclosed employees respond to the loss of privacy. We find that disclosure is associated with an elevated probability that the worker separates from their job in subsequent years. As the employer can only derive a negligible privacy-related benefit from this separation, we interpret this finding as evidence that the incidence of these disclosure costs falls primarily upon nonprofit employees, rather than their employers.

Our results have important implications for our understanding of salary disclosure policies. When paired with an estimate of the social benefits of salary transparency, our estimate of the private costs of salary disclosure could be used to determine an optimal choice of privacy regime. They may also inform the design of optimal differential privacy algorithms in settings where firms and governments publish data on earnings.

Our estimate of the value of privacy pertains to nonprofit employees. As such, these results may not extend to the full population. Furthermore, our bunching design yields a local estimate of the value of privacy among individuals who make close to \$100,000. Additional research is needed to assess whether our estimate generalizes to individuals employed at for-profit firms, or whose incomes are located farther away from the \$100,000 disclosure notch we study.

Finally, we note that a large share of individuals above the disclosure threshold do not appear to respond to the incentives created by the notch. This may either reflect the presence of optimization frictions, or heterogeneity in the preference for salary confidentiality. Future work is needed to disambiguate between these explanations, as well as to clarify the mechanisms underlying the observed reduction in pay.

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Appendix A Matching Procedure

This appendix describes the procedure used to match nonprofit employees’ tax records to their appearance on Form 990 Part VII. Our observations, derived from tax data, are available at the employee-employer-tax year level. Each observation includes the value of reportable compensation paid to the employee by their focal employer, as well as the cumulative value of reportable compensation this employee receives from organizations related to their focal employer. Since, in theory, a focal nonprofit may be unaware of some compensation its related organization may pay to its employees, we assume that the focal-organization reportable compensation is minimally prone to measurement error, and proceed to match on this variable, rather than rely on total reportable compensation.²¹ Besides this information on focal-organization compensation, we are also able to observe the first 4 characters of the employee’s first and last name. We therefore use five variables – employer identification number, tax year, substrings of the first name and surname, and focal-organization related compensation – to assist in fuzzy-matching Form 990 Part VII data to tax data.

When a nonprofit fills out Form 990 Part VII, they must enter an employee’s name and title into Column A. These strings must be parsed and cleaned in order to facilitate the match to tax data. To parse these data, we pre-process the data by removing common prefixes or suffixes, such as “Ph.D.” or “Mrs.” Having cleaned the names, we next apply the *peopleparser* package in R (Lecy, 2020). While this package is successful in the majority of cases, it may fail if a name is improperly digitized. Furthermore, to the extent that nonprofits enter nicknames rather than legal first names, or include maiden names along with current legal surnames, the package may produce a name which fails to match the one recorded in tax data for the corresponding year.

The fuzzy match procedure occurs in four steps. At each step, a successful match between tax data and Form 990 data must match exactly on employer and tax year. The first pass requires that the two sources match exactly on both name and focal-organization reportable compensation. The second pass requires an exact match on name, but allows a fuzzy match on compensation. In the interest of preventing Type I errors, the difference between compensation observed in tax data and compensation reported on the Form 990 is severely curtailed: only differences of up to

²¹Related-organization reportable compensation may also be much easier to misreport. For example, a nonprofit may forget to subtract off the first \$10,000 of reportable compensation paid by related organizations, and therefore report a substantially different amount on the Form 990 than what would be calculated using administrative data.

\$10 are allowed. The third pass requires that the two sources match exactly on compensation, but allows for fuzzy-matching on name. The Levenstein distance between each pair of name strings is restricted to less than 2 in order to generate a match. This restriction is placed for both the first-name strings and surname strings. The fourth and final pass allows for fuzzy-matching on both dimensions.

Figure A1: Probability of a Match between Tax Data and Form 990 Disclosures, by Income and Employer Type

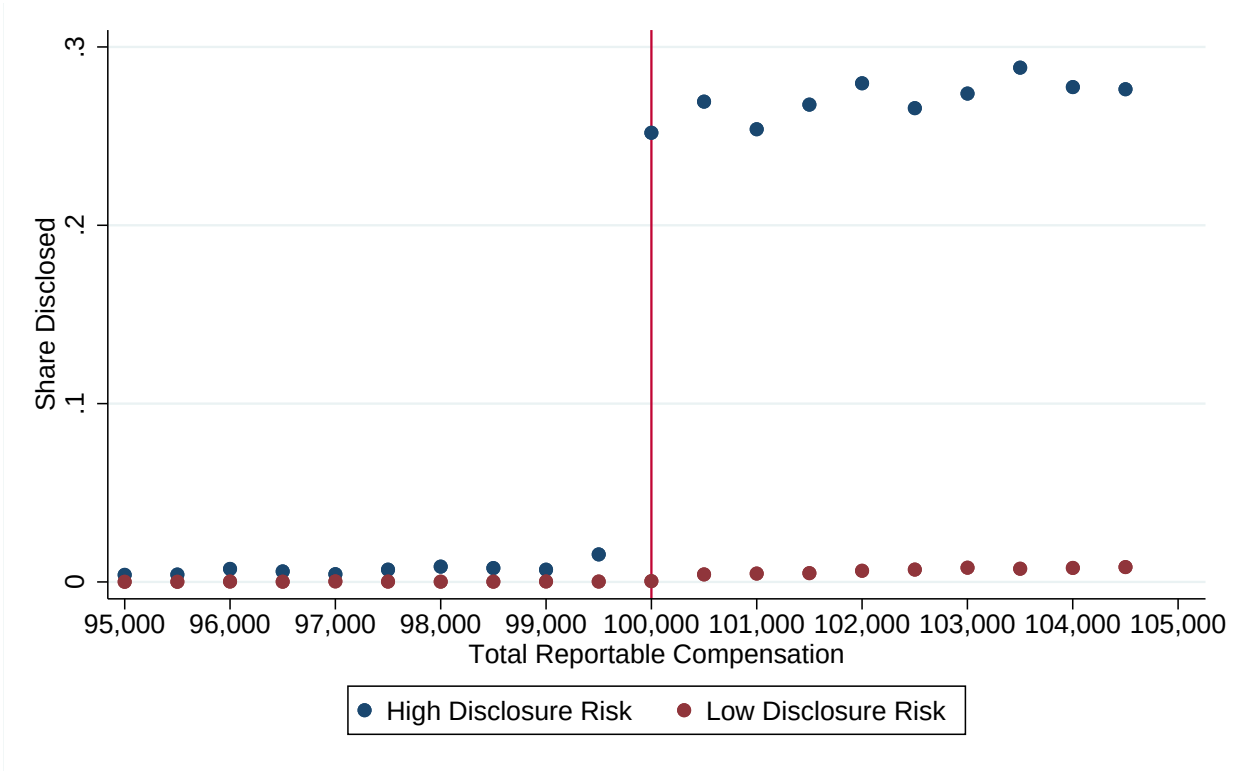
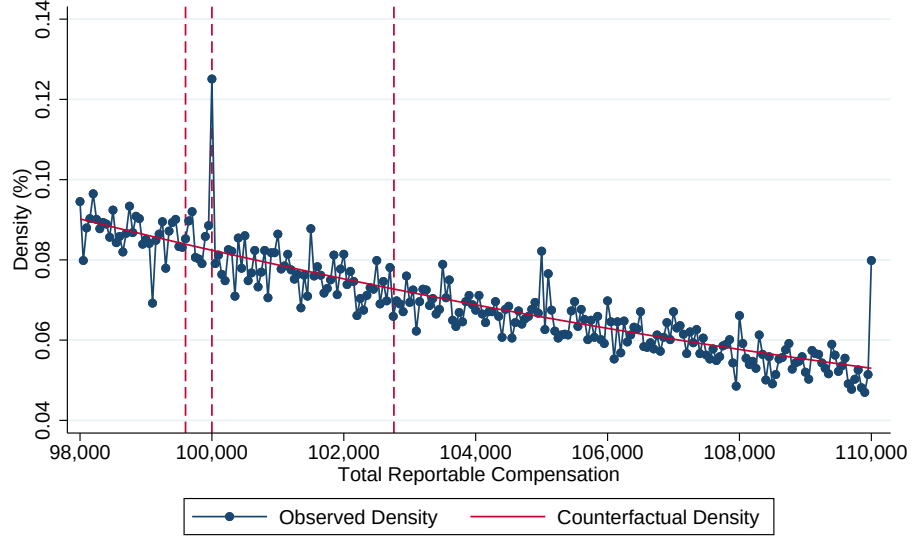


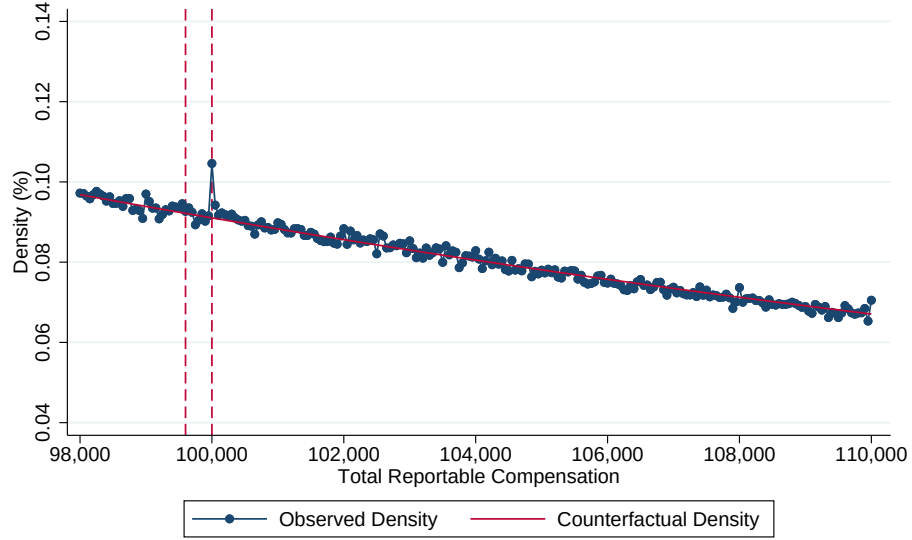
Figure A1 depicts the share of individuals within each bin of total reportable compensation who are successfully matched to the Form 990 Part VII. As all matched individuals are disclosed by definition, this represents both the probability of a match and the probability of disclosure conditional on matching. By comparing this figure with Figure 3, and applying Bayes' Rule, we can characterize the probability of being matched conditional on disclosure as approximately 28%. This match probability pertains to employees of high-disclosure-risk nonprofits, as employees of low-disclosure-risk nonprofits are not expected to appear on Form 990 Part VII at this level of earnings.

Appendix B Robustness Checks

Figure B1: Empirical and Counterfactual Densities Around the Disclosure Notch, using Third-Degree Polynomial



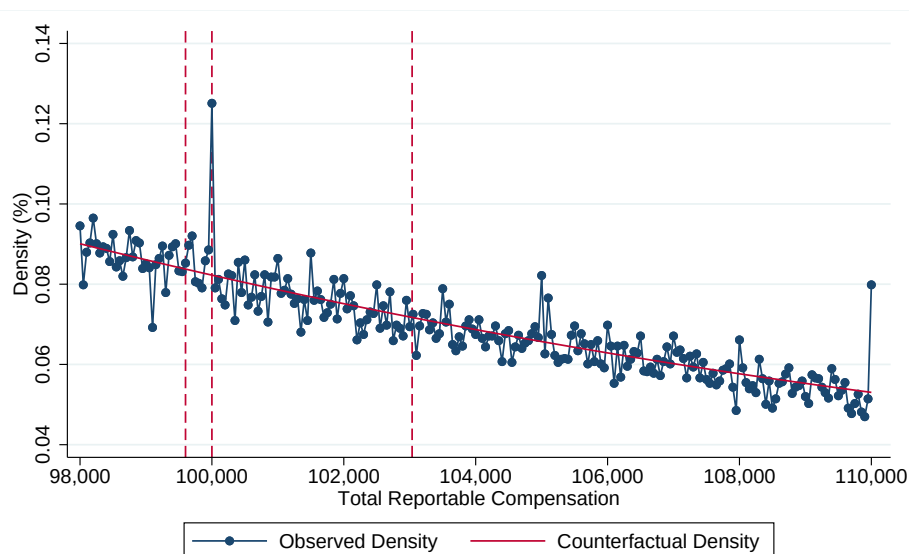
(a) High Disclosure Risk



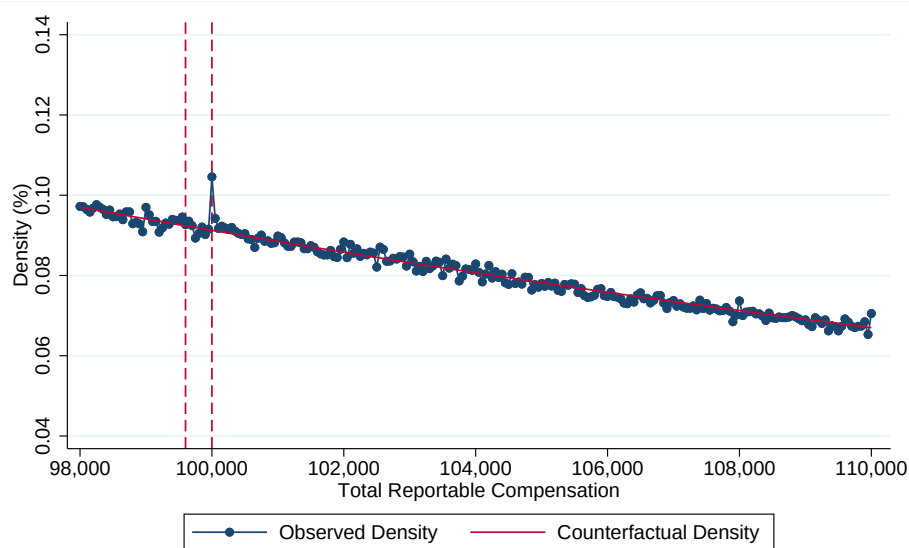
(b) Low Disclosure Risk

Notes: These figures show the observed and counterfactual densities in the neighborhood of the disclosure threshold, as a function of total reportable compensation in current US dollars. Figure B1a shows the actual and counterfactual densities for employees at high-disclosure-risk nonprofits, which have fewer than 5 non-executives listed on Form 990 Section VII. Figure B1b presents these densities for low-disclosure-risk nonprofits, which have exactly 5 non-executives listed on Form 990 Section VII. In each figure, the three red vertical lines denote the lower bound of the excluded range, the disclosure threshold, and the upper bound of the excluded range, respectively. Each point represents a bin with length \$50.

Figure B2: Empirical and Counterfactual Densities Around the Disclosure Notch, using Fourth-Degree Polynomial



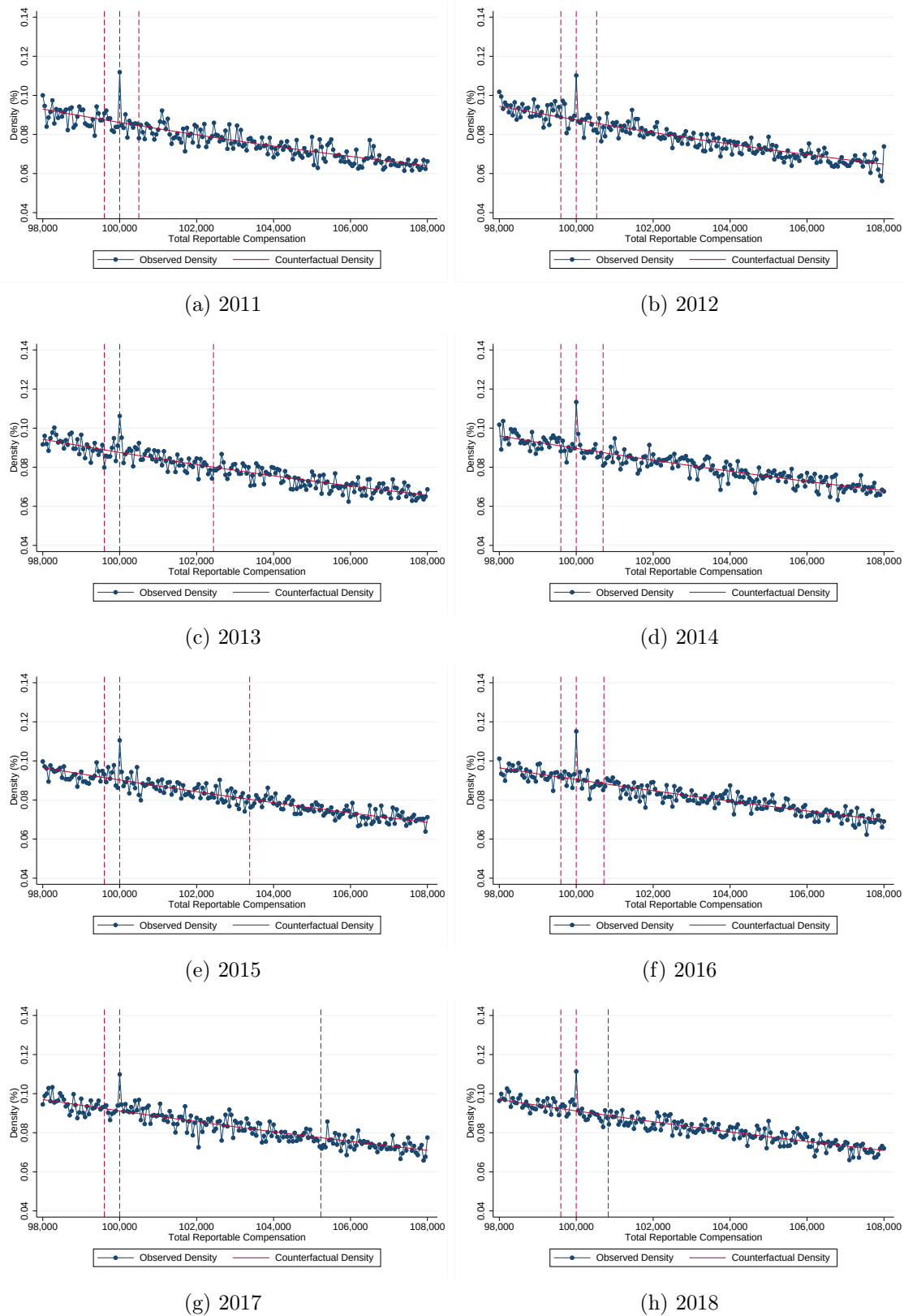
(a) High Disclosure Risk



(b) Low Disclosure Risk

Notes: These figures show the observed and counterfactual densities in the neighborhood of the disclosure threshold, as a function of total reportable compensation in current US dollars. Figure B2a shows the actual and counterfactual densities for employees at high-disclosure-risk nonprofits, which have fewer than 5 non-executives listed on Form 990 Section VII. Figure B2b presents these densities for low-disclosure-risk nonprofits, which have exactly 5 non-executives listed on Form 990 Section VII. In each figure, the three red vertical lines denote the lower bound of the excluded range, the disclosure threshold, and the upper bound of the excluded range, respectively. Each point represents a bin with length \$50.

Figure B3: Empirical and Counterfactual Densities Around the Disclosure Notch, by Year



Notes: These figures show the observed and counterfactual densities in the neighborhood of the disclosure threshold, as a function of total reportable compensation in current US dollars, by year.